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RESEARCH ARTICLE

Ant sharing by plant species bearing extrafloral nectaries has a low impact on plant herbivory in a tropical system

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Abstract

1. Plant species bearing extrafloral nectaries (EFNs) may indirectly influence other plant species by sharing protective ants, potentially altering herbivory levels. However, the propagation of indirect effects in this type of mutualism has seldom been investigated.
2. We investigated indirect effects via ant sharing among 21 EFN-bearing plant species of the tribe Bignonieae distributed on 28 plots in the central Brazilian Amazon. We used an ecological network index to quantify potential indirect effects among plant species through shared dominant and subordinate ant species.
3. Plant species that were more attractive to ants (promoter species) had the highest potential to indirectly influence ant visitation to less attractive plant species (receptor species) in the community, primarily through changes in the attraction of dominant ant species. However, these potential indirect effects did not result in significant differences in herbivory patterns among the studied plants. Ant attendance and herbivory levels were similar among promoter species, neighbouring plants, and non-neighbouring plants. Unlike other mutualisms involving plants (e.g., pollination), the indirect effects among plant species that share protective ants had limited consequences. The low levels of herbivory and the small foraging areas of shared ant defenders could be responsible for the observed limited impact of indirect interactions among plant species in this system.
4. *Synthesis*: This study demonstrates that plant species with EFNs can indirectly influence other plant species through ant sharing. The attractiveness of certain plants to ants enables them to affect the ant visitation patterns of less attractive plants in the community. However, these indirect effects did not significantly alter herbivory levels among plants. Overall, this study advances ecological understanding by showing that the sharing of protective ants among plant species can in some cases confer limited costs and benefits, leading indirect interactions to have minimal impact on herbivory patterns.

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KEYWORDS
ant–plant mutualism, competition, facilitation, herbivory, indirect interactions, Müller's index, mutualistic networks, protection

1 | INTRODUCTION

Diverse mutualistic interactions among organisms involve shared partners across species (Schleuning et al., 2012). One consequence is that species can influence each other indirectly via shared mutualistic partners (Carvalheiro et al., 2014). The magnitude of indirect effects depends on factors such as the number, identity, and functional traits of mutualistic partners shared among different species within a community (Bergamo et al., 2017; Muchhala et al., 2014; Wootton, 1994). It has been widely recognized that these indirect effects can play an important role in the dynamics of natural systems by altering the distribution and abundance of species through complex network interactions within a community (Carvalheiro et al., 2014; Pires et al., 2020; Wootton, 1994; Zakardjian et al., 2020). However, most studies of this phenomenon have focused on pollination mutualisms (but see Barton & Ives, 2014; Guimarães et al., 2017; Martínez et al., 2014; Menge, 1995). This narrow focus limits our understanding of the forces driving patterns of interaction via shared partners and the general mechanisms regulating these indirect effects on mutualisms.

In plant–animal mutualisms, the outcome of indirect interactions between plant species via shared partners can be positive, negative, or neutral (Bergamo et al., 2021). For example, positive indirect effects occur when individuals of a more attractive plant species facilitate the visitation of shared seed dispersers and pollinators to less attractive plant species (e.g., Carlo, 2005; Ghazoul, 2006). Negative indirect effects emerge when the more attractive plant species monopolizes mutualists, reducing the chances that less attractive species will be pollinated or have their seeds dispersed (see Carlo et al., 2003; Rathcke, 1983). Finally, neutral effects are expected whenever sharing mutualistic partners results in neither benefits nor costs (Tur et al., 2016). These possible outcomes are not mutually exclusive across plant species in a community. For example, a given plant species might facilitate mutualist visitation to some species while competing with other species for partners (Vilà et al., 2009). Here, we contribute to the growing body of research on the nature of indirect effects among plant species visited by protective ants, providing novel insights into this poorly studied phenomenon.

Most plants that engage in protective mutualisms with ants exhibit extrafloral nectaries (EFNs), organs that secrete sugar-rich EFN consumed by ants (Bentley, 1977). The ant–plant interactions that result can reduce herbivory and increase plant fitness (Rosumek et al., 2009; Trager et al., 2010). Within the community, these mutualisms are fairly generalized, promoting multiple interactions between individuals of one plant species and individuals of several ant species, leading to a high degree of ant sharing (Rico-Gray & Oliveira, 2007). Consequently, the attractiveness of plants to ants becomes crucial. Attractiveness is influenced by EFN traits, such as size, density, EFN chemistry, and secretion rate (Apple & Feener, 2001; Bentley, 1977). Plants with more EFNs and larger nectar volumes tend to be more attractive to ants (Apple & Feener, 2001; Nogueira et al., 2020). Ants may focus their visits on plant species with the most attractive EFNs (Apple & Feener, 2001), leading less attractive plants to receive fewer ant visits and less protection in their presence. Alternatively, attractive plants may indirectly benefit their less attractive neighbours by enhancing ant abundance and diversity nearby. These alternative outcomes suggest the existence of two categories of plant species: "indirect effect promoter" species (species that promote indirect effects because they are relatively more attractive to ants) and "indirect effect receptor" species (species that receive indirect effects because they are relatively less attractive). We subsequently refer to these as "promoter species" and "receptor species", respectively (termed 'acting' and 'target' species in Bergamo et al., 2021).

The strength of indirect effects on ant–plant interactions should also depend on the traits of the local ant assemblage attracted to EFN-bearing plants. Ant communities are hierarchically structured, with more aggressive and dominant species of higher competitive ability strongly influencing the foraging patterns of subordinate ant species (Parr, 2008). Hierarchical ant dominance is consequently regulated by ant species' abundance and behaviour, with dominant ants exhibiting greater aggressiveness and more effective recruitment for resource exploitation and defence. Because of these features, dominant ants are generally high-quality defenders, defending EFN-bearing plants more effectively than subordinate species (Ness et al., 2006; Xu & Chen, 2010), and tending to monopolize plants with higher-quality EFN (Apple & Feener, 2001; Flores-Flores et al., 2018; Parr, 2008). The interaction among plants sharing ants can lead to two scenarios: competition among plants for ant visitation, in which promoter species reduce visits to receptor species, and facilitation among plants, in which promoter species increase visits to neighbour's receptor species. Both scenarios result in the strongest indirect effects between neighbouring EFN-bearing plants (Seifan et al., 2014; Yang et al., 2011) due to ants' limited foraging area (Baccaro & Ferraz, 2013; Lanan, 2014). However, although similar community-scale dynamics have been examined in other mutualisms, they have rarely been tested in the mutualisms between EFN-bearing plants and ants.

In this study, we investigated the extent to which EFN-bearing plant species propagate indirect effects via shared protective ants, impacting the herbivory patterns between each other at the community level. Specifically, we tested whether competition and/or facilitation occurs between plant species that promote and receive indirect effects. Our general hypothesis is that the presence of promoter plant species in the community affects the frequency of ant visitation and, consequently, the level of herbivory experienced by receptor plant species. This effect can be negative (competition hypothesis) or positive (facilitation hypothesis), depending on the

interactions between EFN-bearing plants and the local ant fauna. Thus, our expectations are as follows: (1a) Promoter plant species, which have larger and more abundant EFNs, will negatively influence the frequency of ant visitation to neighbouring receptor species, those with smaller and less abundant EFNs. Consequently, (1b) receptor plant species will be more heavily attacked by herbivores than promoter plants in the same community (competition scenario). This influence will be especially strong when the promoter species reduces the probability of dominant ants visiting the receptor species. Alternatively, (2a) promoter plant species will increase dominant ant species visitation to receptor neighbour species, and consequently, (2b) receptor plant species will experience lower herbivory than promoter plants in the same community (facilitation scenario).

2 | MATERIALS AND METHODS

2.1 | Study area and focal plant species

We carried out this study at the Reserva Florestal Adolpho Ducke (Ducke Reserve), located 26 km northwest of Manaus, Amazonas, Brazil. Ducke Reserve has an upland tropical forest with a closed canopy and rich understory plant species. The soils form a continuum of clayey latosols, becoming sandier at lower altitudes (Chauvel et al., 1987).

We studied the tribe Bignonieae (Bignoniaceae) (Figure 1), the most abundant and diverse clade of neotropical lianas in the Amazon Forest, which exhibits wide variation in the distribution of EFNs among vegetative units (Lohmann, 2006). The variation in EFN distribution among different species within the tribe creates a gradient of attractiveness for ants (Nogueira et al., 2020). Some species possess highly concentrated clusters of EFNs, eliciting strong ant attraction, while others have sparser or less abundant EFNs, resulting in comparatively lower ant visitation rates (Nogueira et al., 2020).

2.2 | Sampling of ant species feeding on EFN-bearing plants

We established 28 plots of 50 × 10 m (500 m^2 each) distributed in a 9 km^2 grid, separated by at least 500 m. Within each plot, we identified and tagged all juvenile Bignonieae in the forest understory and monitored those actively secreting EFN over time. We focused on juveniles because they are the most visited by herbivores and



FIGURE 1 Interactions between ants and the two plant species with the highest promoter degree across Bignonieae species: (a) *Adenocalymma longilineum*; (b) ants feeding on the extrafloral nectaries on the active nodes of *A. longilineum*; (c) *Adenocalymma moringifolium*; (d) ants from the genus *Crematogaster* feeding on the extrafloral nectaries on the active nodes of *A. moringifolium*.

exhibit the strongest vulnerability to herbivore attacks. Additionally, monitoring juvenile individuals allowed us to track leaf herbivory throughout our sampling consistently. We also mapped the exact spatial location of each plant (latitude and longitude) within each plot.

Extrafloral nectaries of Bignonieae are active only when the plants produce new vegetative tissues (Nogueira et al., 2020). To maximize the chance of detecting interactions with ants, we conducted three censuses (December 2012, February 2013, and May 2013) during the 5 months when plants produced new vegetative tissues. At each sampling period, we performed plant censuses in each plot between 8:00 a.m. and 5:00 p.m., estimating ant visitation to EFNs during 10-min censuses of each plant. After counting ant individuals visiting EFNs on each marked plant, we collected 3–4 workers of each ant morphotype observed on the plants. These were stored in 70% alcohol and later identified to the lowest possible taxonomic level.

After identifying ant species, we assembled a quantitative matrix of the observed frequency of visits of each ant species to each plant species, pooling the observations from the three sampling rounds.

2.3 | Classification of the dominance hierarchy of ant species associated with the plants

Ant species can be classified as dominant and subordinate according to the species' numerical and behavioural dominance concerning resource use (Arnan et al., 2011; Savolainen & Vepsäläinen, 1998). This classification has been commonly linked to the quality of the protection service provided by different ant species to plants bearing EFNs (Leal et al., 2023). In November 2012 and February and May 2013, we observed ant behaviour by placing honey baits on 10 understory plants per plot to identify dominant and subordinate ant species. We created the baits by diluting ~1 mL of honey in 50% water and placing it in a 2 mL Eppendorf tube on the most apical plant part of a stem, close to the leaves, and ensuring that they were the only sugar source available to the ants. After 80–90 min, we counted the number of workers of each ant species on the baits. We preserved them in 70% alcohol for subsequent identification to the lowest possible taxonomic level (Nogueira et al., 2020). We used the same morphospecies codes as Baccaro et al. (2010) and compared the ants to a reference collection from Universidade Federal do Amazonas (UFAM).

Using these data, we quantified the proportion of baits monopolized by each ant species. First, we created an abundance curve based on the proportion of honey baits monopolized by each ant species, a common proxy for ant numerical dominance. The relatively higher efficiency of dominant ants in defending plants with EFNs is not due only to their ability to monopolize plants based on their high numbers, but also to their aggressiveness in defending resources (Leal et al., 2023). For most ant species, numerical and behavioural dominance are correlated. For this reason, to classify the ants as dominants or subordinates, we complemented our data on the proportion of baits monopolized by each ant species with information from Baccaro et al. (2010), who evaluated the behavioural dominance of the same set of ant species at our study sites using competitive arena experiments. We ranked the ant species according to the proportion of honey baits they monopolized. The ant species with the lowest proportion of monopolized baits but which was classified as a behavioural dominant by Baccaro et al. (2010) was used as the rank threshold. We classified the ant species above this threshold as dominant and those below as subordinate. Thus, our categorical classification groups ant species based not only on data on numerical monopolization of resources, but also on their ability to defend food resources.

2.4 | Indirect effects across plant species via shared protective ants

Using the matrix of visitation frequencies of ant species to each EFN-bearing plant species, we created a plant–plant species matrix with values representing potential indirect effects between co-occurring plant species. We estimated these potential indirect effects using a network index proposed by Müller et al. (1999) (hereafter, Müller's index). This index has been used as a proxy of potential for apparent competition between species sharing resources, whether in antagonistic or mutualistic interactions (Bergamo et al., 2017, 2021; Carvalheiro et al., 2014; Müller et al., 1999; Tack et al., 2011). In our case, it represented how much a plant species (promoter species) contributes to the frequency of ant visitation to other co-occurring plant species (receptor species).

We calculated Müller's index with the PAC function of the *bipartite* R package using the following formula:

$$d_{ij} = \sum_{n}\left[\frac{\propto_{ik}}{\sum_{l}\propto_{il}}\frac{\propto_{jk}}{\sum_{m}\propto_{mk}}\right],$$

where d_{ij} corresponds to the potential indirect effect of plant species j on plant species i; $\sum_n$ is the sum of all n ant species shared between plant species i and j; α_{ik} corresponds to the visitation frequency of shared ant k to plant i; α_{ik} is the visitation frequency of shared ant k to plant j; $\sum_l \alpha_{il}$ corresponds to the total number of visits that the plant i received from all l ants in the network and; $\sum_m \alpha_{mk}$ is the total number of visits by the shared ant k to all m plants in the graph. We restricted the index calculation to plant species pairs with EFNs simultaneously active in the plots since only plants secreting EFN can indirectly influence each other through ant sharing. We excluded plants with no active EFNs from the matrix prior to Müller's index calculations.

Müller's index ranges from 0 when a given plant species does not affect ant visitation to another plant, to 1, representing the maximum effect of one plant species on patterns of ant attendance to another plant species. Thus, d_{ij} represents how much j affects i. The column values in the pairwise plant–plant matrix represent how much one plant species indirectly affects other plant species. The rows represent how much a given plant species is influenced by other co-occurring plant species. The index is asymmetric and describes the expected indirect effects between different plant

species within the same community. Because the index values are calculated between plant species pairs, the same plant species can have different Müller's index values depending on the plant species to which it is being compared.

Given that we expected dominant ant species to have greater potential to generate indirect effects among plant species compared to subordinate ants, we decomposed the indirect effects between the plant species pairs (i.e., Müller's index) according to the ant dominance hierarchy. This procedure involved decomposing Müller's index of plant species *j* into two values: one for cases in which plant species shared dominant ant species and another for cases in which they shared subordinate ant species.

We used values of Müller's index to create quantitative ant-overlap graphs. These graphs visually represent the strength of indirect effects through shared ants among Bignonieae species with active EFNs in our plots. They illustrate the importance of a given plant species as a resource for all ants shared with the other EFN-bearing plant species. The plant species are represented as vertices (dots of different sizes in Figure 2), and their size is proportional to the number of indirect links (edges) each plant species has with other plant species through shared protective ants, indicating their importance for ants. The width of the link at each end connecting two plant species represents the magnitude of one plant species' influence on another (Müller et al., 1999). Since the indirect effects of one plant species on another are asymmetric, the ends of the link between two plant species have different widths. We built three graphs to represent potential indirect effects between plant species based on our hypotheses: one graph illustrates the sharing of all ant species, while the other two illustrate the sharing of the dominant or subordinate ant species only.

We used Bergamo et al.'s (2021) approach to obtain global Müller's index values per plant species, representing the role of each plant species on the potential indirect effects within a community. We obtained the promoter degree (referred to as 'acting degree' in Bergamo et al., 2021) of each plant species by summing Müller's index column values for each plant species. The promoter degree indicates the species used as a food resource by all ants shared with other EFN-bearing plant species. We obtained the receptor degree (referred to as 'target degree' in Bergamo et al., 2021) by summing Müller's index row values for each plant species. The receptor degree indicates how much a plant species is affected by other species through shared ants. After estimating the promoter and receptor degree of each plant species when sharing all ant species available, we used the same procedure to calculate the plant species promoter and receptor degree when sharing only dominant or subordinate ant species.

Müller's index is proportional to the total visitation each plant species received in the network. Thus, it is possible that both heavily and poorly visited plants exhibit similar promoter degree values, even though they experience different magnitudes of interactions. To better understand this relationship, we looked at the relationship between the promoter degree and the number of interactions which each plant species. There was a strong positive correlation between the promoter degree and the total number of visits for all ants ($r=0.88$; $p<0.001$) and dominant ants ($r=0.92$; $p<0.001$), but not for subordinate ones ($r=0.16$; $p=0.49$). Thus, to some extent, the promoter degree also reflects the magnitude of interactions, especially when considering dominant ants.

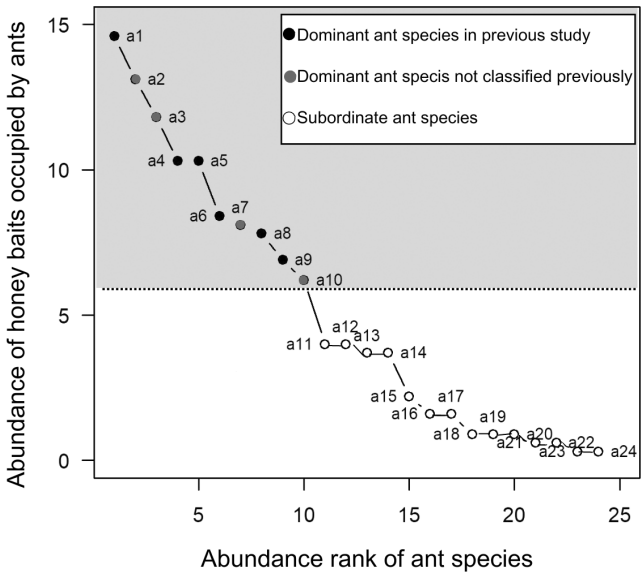


FIGURE 2 We plotted a rank abundance curve of occupied baits for each ant species that feed on the extrafloral nectaries of the Bignonieae plant species and overlaid the dominant and subordinate ant species classified by a previous study (Baccaro et al., 2010). The dominant ant species were represented by the grey background in the graph, while the subordinate ants were represented by the white background. The codes correspond to the following ant species: a1 = *Crematogaster brasiliensis*; a2 = *Nylanderia* sp.1; a3 = *Pheidole* sp.25; a4 = *Crematogaster limata*; a5 = *Pheidole* sp13; a6 = *Ochetomyrmex semipolitus*; a7 = *Crematogaster sotobosque*; a8 = *Wasmannia auropunctata*; a9 = *Crematogaster tenuicula*; a10 = *Crematogaster carinata*; a11 = *Crematogaster femoratus*; a12 = *Ochetomyrmex neopolitus*; a13 = *Brachymyrmex* sp1; a14 = *Solenopsis* sp.2; a15 = *Azteca* sp.2; a16 = *Megalomyrmex goeldii*; a17 = *Pheidole* sp.100; a18 = *Neoponera unidentata*; a19 = *Pheidole* sp.4; a20 = *Paratrachymyrmex bugionini*; a21 = *Dolichoderus bispinosus*; a22 = *Solenopsis* sp5; a23 = *Crematogaster longispina*; a24 = *Ectatomma tuberculatum*.

2.5 | Describing the EFNs traits across plant species

Given that plant species vary in attractiveness to ants, we quantified the plant investment in EFNs to investigate if and how it might determine the role of each plant species as a promoter or receptor of indirect effects within a community.

Bignonieae species have glandular trichomes that secrete EFN (Nogueira et al., 2013). These trichomes are typically found on certain portions of the stem, axillary bud prophylls, and leaves (Nogueira et al., 2012), which are repeated at each stem node or module of the vegetative portion of the plant (Nogueira et al., 2012, 2020).

We estimated plant species' attractiveness to ants by assessing two traits: (1) the secretory area of EFNs at each node of the plant stem and (2) the number of nodes with clusters of active EFNs that ants frequently visit. To calculate the first metric, we multiplied the average size of each secretory trichome's head by the total number of secretory trichomes area in different plant parts. We measured the second trait according to Nogueira et al. (2020), who showed that the number of recently formed shoot nodes positively correlates with ant attendance in the same Bignonieae plant species evaluated here (Nogueira et al., 2020). However, other factors could contribute to plant attractiveness, including the chemical composition of EFN, which has yet to be investigated in this system.

2.6 | Characterization of leaf herbivory

To assess the impact of co-occurring plants with EFNs on each other's ant attendance and the resulting leaf damage, we measured leaf herbivory levels on each focal plant in our plots during the final field census in May 2013. We assessed herbivory on each plant by collecting at least 20% of the fully expanded leaves and separating the leaflets from the leaves for drying. We photographed the dried leaflets against a black background containing a known scale (3 cm × 3 cm). We then used *ImageJ* software to estimate the percentage of each leaflet consumed by herbivores. We calculated the median percentage of the damaged area per leaflet to account for non-normal distributions. Additionally, we assessed the proportion of leaflets that experienced a loss of more than 5% of their area. This 5% threshold was based on the recognition that an herbivory threshold exists (see Trumble et al., 1993) below which damage to leaf performance is expected to have minimal impact on plant fitness.

Two species had numerous tiny leaflets (>50 leaflets/leaf). To ensure a representative sample for herbivory descriptors for these species, we subsampled leaflets from each plant. We divided the photographs of these species into a 2 × 2 grid with 7 × 9 cm in each area. We then separated each of the four areas into 2 cm × 2 cm squares to systematically pick leaflets along each area. For individuals with 50–100 leaflets per leaf, we sampled 20 leaflets; for individuals with more than 100 leaflets per leaf, we sampled 40. Finally, to estimate the herbivory descriptors per plant species, we calculated (i) the median of the herbivory percentage across plants and (ii) the average proportion of damaged leaflets across plants.

2.7 | Data analysis

Our first hypothesis was that investment into EFN traits would determine species' roles as promoters and receptors of indirect effects within a community. To evaluate this hypothesis, we modelled plant species' promoter and receptor degrees, describing the potential indirect effects as response variables and the two EFN traits (number of active nodes and the secretory area of EFNs) as fixed predictors. We utilized a two-step analytical approach in this analysis. First, we used a model selection method to determine the variance structure as a function of predictors in Generalized Least Squares (GLS) models (Zuur et al., 2009). We fitted a total of six models, with three models using distinct promoter degree as response variables—global promoter degree and promoter degree partitioned by dominant and subordinate ant species—and three other models for distinct receptor degree as response variables—global receptor degree and receptor degree when sharing dominant and subordinate ant species. We present these models' syntax and the results of GLS model selection in Tables S1–S3. Subsequently, we performed a variance inflation factor (VIF) analysis for each selected model to detect multicollinearity between the two fixed predictors (EFN traits). The results revealed no multicollinearity (<3) between the predictors of all selected models. Therefore, we treated each model as a multiple regression and assessed the *p*-values associated with each predictor using maximum likelihood. This procedure allowed us to determine whether one or both predictors describing EFN traits were related to potential indirect effects across plant species in the community.

Regarding herbivory, we expected opposite results for the two hypotheses. Our *competition hypothesis* would be supported if plant species with higher receptor degree exhibit greater herbivory than species with higher promoter degree. Alternatively, the *facilitation hypothesis* would be supported if plant species with higher receptor degree exhibit lower herbivory than plant species with higher promoter degree. To test these predictions, we used general linear models, with the average percentage of leaf herbivory or proportion of leaflets consumed per plant species as response variables and the species promoter and receptor degrees as fixed predictors. We applied a square root transformation to the percentage of leaf herbivory data to ensure the residuals conform to the normality assumption. VIF analyses did not show multicollinearity between the predictors (promoter degree and receptor degree), either considering the entire ant assemblage or the partitioned dominant and subordinate ant sharing (VIF values <3 for all variables).

2.8 | Spatial structure analysis

The closer in space two plants are to each other, the higher the probability they will share the same ant partners and indirectly influence each other's ant visitation patterns. However, Müller's index does not explicitly account for spatial distance between plants. To overcome this limitation, we conducted an explicit spatial analysis. Since plant species composition varied among plots, we identified the species with the highest promoter degree value in each plot and established a circular area with a 5-m radius around these plants. We classified plants within this range as 'neighbour plants', those we expected to be under the strongest indirect effects of the promoter plant species, and those outside that range as 'non-neighbour plants'. We opted for a 5-m radius based on a prior investigation conducted in our study area, which indicated that most dominant ant species seldom forage further than 5 m from their nests (Baccaro & Ferraz, 2013). To account for potential ant colonies located between

two focal plants, we sampled a circular area with a radius twice as large as the estimated foraging distance. We only included plots with at least two individuals of different plant species spaced at least 10 m apart. In some plots, we observed no plant species other than the focal promoter species within this 5-m radius area, preventing evaluation of the indirect effect of promoter species. For this reason, we excluded these plots from this analysis, which reduced our dataset from 28 to 12 plots (83 individual plants).

After identifying neighbour and non-neighbour plants, we investigated differences in the probability of dominant and subordinate ant occurrence on plants (presence/absence) and the variation in the abundance of these ants on the plants. We also evaluated variation in leaf herbivory among promoter plant species, their neighbours, and non-neighbour plants within each plot. To accomplish this, we used Generalized Linear Mixed Models (GLMMs) with a binomial random distribution for ant occurrence, a GLMM with a binomial negative random distribution for ant abundance, and a Linear Mixed Model with a Gaussian random distribution for mean leaf herbivory. In the case of leaf herbivory, we applied an arcsine transformation before analysis.

2.9 | Random factors and evaluation of fixed effects

For all models, we set the plot and plant species identity in all models as random variables. For the fixed effects in each model, we computed the standardized effect sizes (b_{st}) following Cohen (1988) along with 95% confidence intervals. We performed all analyses in the software R 4.1.2 (R Development Core Team, 2021), using lme4 (Bates et al., 2015), nlme (Pinheiro et al., 2022), and glmmTMB (Brooks et al., 2017) packages.

3 | RESULTS

We studied 198 plant individuals from 21 Bignonieae species with active nodes in 28 plots. *Adenocalymma longilineum* was the most abundant plant species, with 54 individuals (27.3% of all plants) exhibiting active nodes. We recorded 909 workers from 32 ant species visiting the plants, with *Crematogaster tenuicula* (202 interactions; 22.2% of all records) being the most frequent, followed by *C. limata* (173 interactions; 19.03%) and *C. brasiliensis* (157 interactions; 17.3%). On average, each plant had 4.7 ± 10.2 (mean ± standard deviation) ant workers during each 10-min census.

3.1 | Dominance hierarchy of ant species associated with the plants

Before exploring the potential indirect effects among plants mediated by shared ants, we examined the dominance hierarchy across ant species visiting our focal EFN-bearing plant species. Among the 837 baits in our 28 plots, ants visited 323 during the three sampling periods. Among the 32 ant species that visited the EFNs, 24 also visited the baits, and eight were rare in our plants. In all sampling periods, each bait was monopolized by a single ant species. *Crematogaster brasiliensis* was the most frequent (14.6% of baits), followed by *Nylanderia* sp.1 and *Pheidole* sp.25 (13.1% and 11.8% of the baits, respectively).

We classified the ant species with the 10 highest proportions of monopolized baits as dominant. Six of these were also classified as dominant by Baccaro et al. (2010); the other four species were not evaluated in that study (Figure 2). However, data available in the literature show that these four species exhibit aggressive behaviour, which enables them to competitively exclude other ant species from food resources (e.g., Erickson et al., 2021; F. Baccaro, personal observation). For these reasons, we also classified these four species as dominants. Furthermore, given the minimal difference in our rank abundance between the last ant species classified as dominant ant species (*Crematogaster tenuicula*) and the next one (*Crematogaster carinata*) (Figure 2), we included *C. carinata* as dominant in our study. Thus, we classified 10 ant species as dominant. The remaining 23 ant species, including the eight that did not visit our baits and the 14 that rarely dominated the baits, were classified as subordinate (Figure 2). Baccaro et al. (2010) surveyed two of these 23 subordinate species and considered both subordinates.

To provide further information about the biological reliability of our ant dominance classification, we performed an additional analysis a posteriori to evaluate the power of our ant categorization in predicting the biology of ants on Bignonieae. If our classification represents the patterns of ant resource use in nature, we expect that dominant ant species recruit more ants to EFNs than subordinate ant species due to dominant ants' ability to numerically and behaviourally monopolize food resources. Therefore, we tested if the ant species classified as dominant recruit more workers, on average, to EFNs (the total number of ants per plant standardized by the number of plant nodes with active EFNs) than the subordinate species. We found that ant species classified as dominant recruited more than three-fold more workers than those classified as subordinate (Figure S1). Finally, dominant ant species were responsible for 69.8% of observed interactions with EFNs. This number corresponds to more than twice the observed interactions between our focal plants and subordinate ant species (30.1%). These data indicated that our categorical classification efficiently captures the biological differences among the ant species visiting our plants, including the focal traits that could modulate the indirect effects of promoters on receptor species.

3.2 | The potential indirect effect between plant species via shared ants

We used three ant-overlap graphs to explore potential indirect effects among plant species mediated by shared ants. The first revealed 19 links among 12 Bignonieae species bearing EFNs (57.2%

of total EFN-bearing plant species), describing the potential indirect effects via shared ants (Figure 3a). In contrast, nine plant species were not linked to any other plant species through shared ant species. This occurred due to one of three reasons: the plant species attracted ants not observed on any other plant species; the plant species did not co-occur; or the plant species attracted no ants (the least common reason for the absence of links between plant species). Two plant species, *A. longilineum* and *A. moringifolium*, were notable as promoters of potential indirect effects on other species, especially via sharing dominant ant species (Figure 3a). *Adenocalymma longilineum* monopolized ant visits on all links with other plant species, and *A. moringifolium* monopolized ants on four of its links. In contrast, *A. adenophorum* did not monopolize ant visits shared with other plant species and instead received most potential indirect effects from other plant species.

The second ant-overlap graph showed 17 links between plant species that shared dominant ant species (Figure 3b). Among the 21 plant species examined, 10 (47.6% of total EFN-bearing plant species) shared dominant ant species with other plant species. *Adenocalymma longilineum* and *A. moringifolium* were especially important in promoting potential indirect effects through shared dominant ant species. As was observed for the entire ant assemblage, *A. adenophorum* did not monopolize shared dominant ants across any link with the other five plant species with which it indirectly interacted. Instead, it received the most potential indirect effects. Notably, the shared dominant ant species generated the most potential indirect effects across plant species, as seen by the similarity between Figure 3a and 3b.

The third ant-overlap graph considered subordinate ant species and revealed only eight links between plant species (Figure 3c), less than half of the links established via dominant ant species. Among the 21 plant species examined, only eight (31.8% of total EFN-bearing plant species) shared subordinate ants with other plant species. *Adenocalymma longilineum* and *A. moringifolium* lost importance in determining the potential indirect effects between plants via the subordinate ant species.

3.3 | Impact of EFN traits on potential indirect effects between plants via dominant and subordinate ant species

We investigated the effect of two EFN traits (number of active nodes and EFN secretory area) on the potential indirect effects (promoting and receptor degree values) across plant species, as we had hypothesized that nectary traits would play a role in promoting or receiving indirect effects in the community. The number of active nodes was positively related to the promoter degree for the entire ant visitor assemblage, dominant ant species, and subordinate ant species (Figure 4a–c). Plant species with a higher number of active nodes had higher values of promoter degree (Table 1). On the other hand, the EFN secretory area was not related to the promoter degree values (Table 1).

The receptor degree of plant species considering all shared ant species did not depend on the number of active nodes and the EFN secretory area (Figure S2a; Table 1). We observed a similar pattern when examining the receptor degree via sharing only dominant ant species (Figure S2b) and subordinate ant species (Figure S2c; Table 1).

3.4 | Impact of indirect effects on leaf herbivory patterns

We described leaf herbivory patterns by measuring the proportion of leaflets with more than 5% of the leaf blade consumed and the average percentage of leaf blade removed. On average, plant species

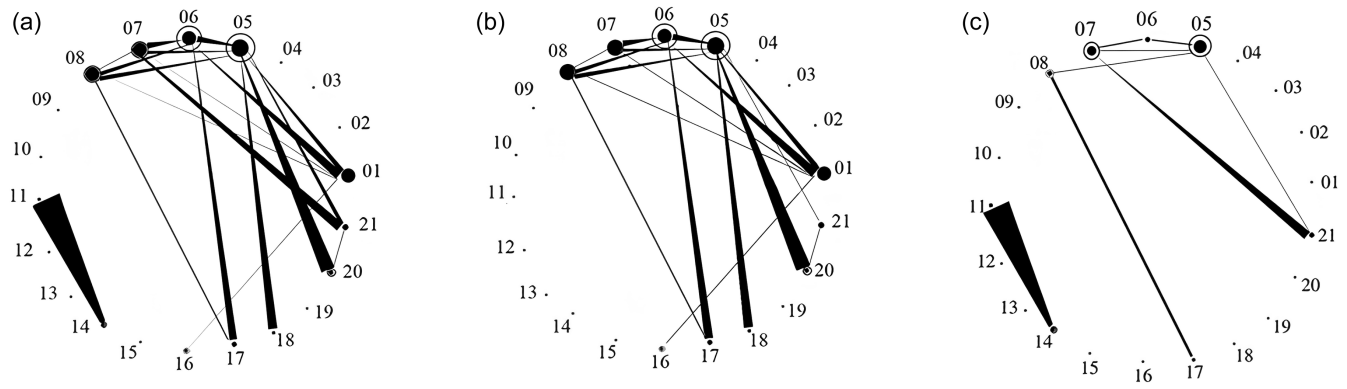


FIGURE 3 Graphs representing potential indirect effects generated by (a) the entire ant assemblage, (b) the dominant ant species, and (c) the subordinate ant species. The figure shows plant species as vertices, with their size representing their importance for ants through shared protective ants. The width of the edges between plant species indicates the influence of one species on another, with different widths reflecting asymmetric indirect effects. The codes correspond to the following plant species: 01—*Adenocalymma adenophorum*; 02—*A. bracteousm*; 03—*A. flaviforum*; 04—*A. japurensis*; 05—*A. longilineum*; 06—*A. moringifolium*; 07—*A. validum*; 08—*A. tanaeciircapum*; 09—*Amphilophium parkeri*; 10—*Anemopaegma robustum*; 11—*Bignonia aequinoctialis*; 12—*B. prieuri*; 13—*Calichlamys latifolia*; 14—*Fridericia trailli*; 15—*F. nigrescens*; 16—*F. prancei*; 17—*F. triplinervia*; 18—*Pachyptera aromatica*; 19—*Pleonotoma albiflora*; 20—*P. melioides*; 21—*Tynanthus panurensis*.

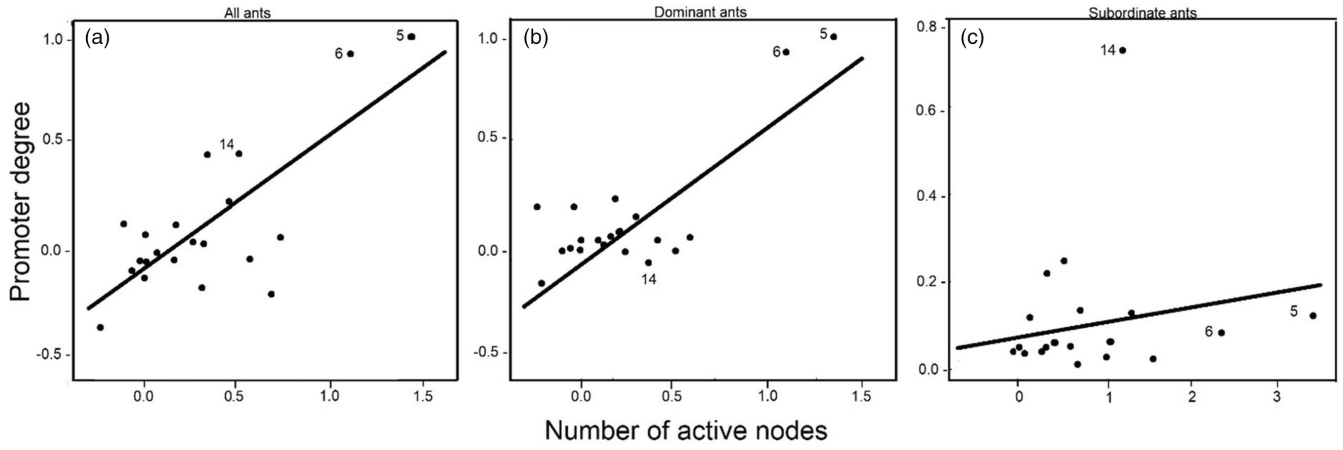


FIGURE 4 General Least Square (GLS) relationship between the number of active nodes and the promoter degree across plant species. (a) More numbers of active nodes were associated with higher promoter degree for plant species across the entire ant assemblage feeding on extrafloral nectaries. (b) This relationship held true when considering the promoter degree of plant species generated by the dominant ants. (c) When considering promoter degree partitioned by the shared subordinate ants, plant species with more active nodes still had higher promoter degree. Partial GLS residual plots show the expected effect of one variable when the other variables are held statistically constant in the model. The partial residual plots include a trendline adjusted by an ordinary linear regression model. Species codes: 5 = *Adenocalymma longilineum*; 6 = *Adenocalymma moringifolium*; 14 = *Fridericia trailli*.

TABLE 1 Relationship between the potential indirect effects (promoter and receptor degrees) and EFN traits (number of active nodes and EFN secretory area).

Response variables (ant assemblage considered)	Fixed factors	$b_{st} \pm$ CI	*t*	*p*	Graph
Promoter degree (all ant species)	Active nodes	0.81 ± 0.96	3.5	**0.002**	Figure 4a
	EFN secretory area	0.0 ± −0.47	0.03	0.97	-
Promoter degree (dominant ant species)	Active nodes	0.60 ± 0.84	3.0	**0.004**	Figure 4b
	EFN secretory area	−0.05 ± −0.4	−0.55	0.58	-
Promoter degree (subordinate ant species)	Active nodes	0.76 ± 1.88	1.69	**0.04**	Figure 4c
	EFN secretory area	0.07 ± 1.05	0.26	0.79	-
Receptor degree (all ant species)	Active nodes	0.23 ± 20.6	−0.8	0.42	Figure S2a
	EFN secretory area	1.19 ± 1.49	0.6	0.53	-
Receptor degree (dominant ant species)	Active nodes	−1.82 ± −7.93	−0.96	0.61	Figure S2b
	EFN secretory area	2.30 ± −12.37	0.78	0.40	-
Receptor degree (subordinate ant species)	Active nodes	0.15 ± 0.06	0.93	0.41	Figure S2c
	EFN secretory area	0.01 ± 0.16	0.16	0.87	-

Note: All tests below will represent the standardized effect size estimate and the 95% confidence intervals ($b_{st} \pm$ 95% CI). Bold values are statistically significant at the 5% level. $N = 21$ plant species.
Abbreviation: EFN, extrafloral nectar.

had 0.56 ± 0.25 (mean ± SD) of the leaflets with more than 5% of the leaf blade consumed. The percentage of leaves consumed was 3.4 ± 11.6% per individual plant and 5.4 ± 12.4% per plant species.

Using estimated herbivory patterns, we tested whether plant species that promote indirect effects by attracting more ants influenced leaf herbivory of plant species that received indirect effects negatively (*competition hypothesis*) or positively (*facilitation hypothesis*). We found no relationship between the proportion of leaflets consumed per plant species and either the promoter or receptor degree, either considering the entire ant assemblage or dominant and subordinate ant species separately (Table 2). Additionally, there was no relationship between the percentage of leaves consumed per species and either the promoter or the receptor degree of the plant species, either for the entire ant assemblage or for dominant and subordinate ant species (Table 2).

3.5 | Local impact of promoter plant species on ant visitation pattern and herbivory on neighbour and non-neighbour plants

We also tested the *competition* and *facilitation hypotheses* at the local scale (i.e., in the vicinity of the promoter plant species). The probability of interaction between plants and dominant and subordinate ant

TABLE 2 Relationship between the potential indirect effects (promoter and receptor degrees) and herbivory patterns (proportion of leaflets consumed and percentage of leaves consumed per species).

Response variables	Fixed factors (ant assemblages)	$b_{st} \pm CI$	t	p	Graph
Proportion of leaflets consumed	Promoter degree (all species)	−0.17 ± −1.02	−0.64	0.52	Figure S3a
	Receptor degree (all species)	0.00 ± −1.01	0.02	0.98	-
	Promoter degree (dominant species)	0.00 ± −1.03	0.03	0.98	Figure S3b
	Receptor degree (dominant species)	−0.22 ± −1.04	−0.84	0.40	-
	Promoter degree (subordinate species)	0.33 ± −1.03	1.24	0.21	-
Percentage of leaves consumed per species	Receptor degree (subordinate species)	−0.34 ± 1.03	−1.31	0.19	-
	Promoter degree (all species)	−0.11 ± −0.61	−0.75	0.45	-
	Receptor degree (all species)	0.04 ± −0.06	0.26	0.79	Figure S3c
	Promoter degree (dominant species)	0.00 ± −0.61	0.00	0.99	-
	Receptor degree (dominant species)	−0.14 ± −0.61	−0.94	0.34	-
	Promoter degree (subordinate species)	0.27 ± −0.59	1.77	0.07	Figure S3d
	Receptor degree (subordinate species)	−0.23 ± −0.59	−1.52	−1.52	-

Note: All tests below will represent the standardized effect size estimate and its 95% confidence intervals ($b_{st} \pm$ 95% CI). $N = 20$ plant species.

TABLE 3 Relationship between the probability of interaction of plants with dominant and subordinate ants, the average number of dominant and subordinate ants visiting EFNs, and the average percentage of leaf herbivory considering the vicinity of the promoter plant species.

Response variables	Fixed factors	$b_{st} \pm CI$	z	p	Graph
Probability of interaction between plants and ants (Local Scale)	Neighbour plants (dominant ants)	−0.66 ± −2.39	−1.07	0.28	Figure S4a
	Neighbour plants (subordinate ants)	−2.23 ± −6.36	−1.37	0.16	Figure S4a
	Non-neighbour plants (dominant ants)	0.00 ± −1.01	0.80	0.42	Figure S4a
Average number of ants visiting EFNs (Local Scale)	Non-neighbour plants (subordinate ants)	−0.40 ± −5.32	−0.29	0.76	Figure S4a
	Neighbour plants (dominant ants)	−1.14 ± −3.86	−1.15	0.24	Figure S4b
	Neighbour plants (subordinate ants)	−0.83 ± −3.31	−0.97	0.32	Figure S4b
	Non-neighbour plants (dominant ants)	1.48 ± −5.22	1.11	0.26	Figure S4b
	Non-neighbour plants (subordinate ants)	−0.26 ± −5.5	−0.18	0.85	Figure S4b
Average percentage of leaf herbivory (Local Scale)	Neighbour plants	−0.57 ± −1.32	−1.69	0.09	Figure S5
	Non-neighbour plants	−0.20 ± −1.74	−0.45	0.65	Figure S5

Note: All tests below will represent the standardized effect size estimate and its 95% confidence intervals ($b_{st} \pm$ 95% CI). $N = 83$ plant individuals.
Abbreviation: EFN, extrafloral nectar.

species did not differ among plant species with higher promoter degree values per plot, their neighbours, and non-neighbour plants (Table 3). On average, the number of dominant and subordinate ants visiting the EFNs was the same in the plant species with higher promoter degree values per plot, their neighbours, and non-neighbour plants (Table 3). At the plot scale, the average percentage of leaf herbivory was 5.1 ± 12.4% per plant. The percentage of leaf herbivory per plant was similar among plant species with the highest promoter degree values per plot, their neighbours, and non-neighbour plants (Table 3).

4 | DISCUSSION

Plant species bearing EFNs share an ant assemblage that can protect against herbivores. Sharing of mutualistic ant species can generate indirect effects among plants. Here, we found that the number of nodes secreting EFN was a key trait defining the role of plant species as a promoter (one exerting an indirect influence) but not as a recipient plant species (one suffering the consequences of such influence) in the community. Contrary to our expectations, however. the extent of herbivory was unrelated to a plant species' promoter and receptor degree. These results show that although some plant species attract relatively more ant species, especially dominant ones, this does not necessarily generate widespread indirect effects across plants that result in modifications to leaf herbivory patterns.

4.1 | Impact of plant traits on potential indirect effects via dominant and subordinate ants

Extrafloral nectar traits have been commonly related to ant visitation patterns and the outcome of protection services they receive in

return (e.g., Nogueira et al., 2015, 2016). Here, EFN traits played a major role in determining which plants function as promoter species, monopolizing ant visits by producing more active nodes with EFNs. The number of active nodes is a reliable indicator of food availability at a small spatial scale (Nogueira et al., 2020). As a result, when interacting with these plants, worker ants can exploit multiple EFN aggregations within the same individual plant, thus reducing foraging costs and increasing net energy gained per unit of nectar consumed (MacArthur & Pianka, 1966; Pyke, 1984). Therefore, it is reasonable that the same trait regulating plant attraction to ants also regulates the chances that a plant species will act as a promoter of indirect effects within a community. Surprisingly, however, none of the EFN traits explored here determined the receptor degree of the plant: lower attractiveness to ants did not translate into being more affected by other plant species. Thus, indirect effects in ant protection mutualisms may be less asymmetrical than the ones observed in other mutualisms, such as pollination (e.g., Bergamo et al., 2021). In this ant-plant protection system, EFNs-bearing plants share many protective ants available in the habitat, regardless of their attractiveness to the ants.

Two plant species with the highest number of active nodes (*A. longilineum* and *A. moriginfolium*; Figure 1a,d) monopolized ant visitation, receiving approximately half of all visits recorded across the 21 plant species. Thus, these two species act as potential promoters of indirect effects in the community. Plants that invest more in EFNs are more likely to be attended by ants (Apple & Feener, 2001; Flores-Flores et al., 2018; Nogueira et al., 2020; Souza et al., 2022). When considering subordinate ant species, the promoter degree was positively correlated with the number of active nodes. However, in this case, *A. longilineum* and *A. moriginfolium*, the most attractive plant species to dominant ant species, were no longer the main species driving the indirect effects on other EFN-bearing plants via the sharing of subordinate ants. This result is likely explained by dominant ants competitively displacing subordinate species from highly rewarding food resources available in the habitat (Cerdá et al., 2013; Parr & Gibb, 2010). Thus, subordinate ant species are more prone to interact with less attractive plant species. This result highlights an important implication for the dynamics of indirect effects among partners in different mutualistic systems: the magnitude of indirect effects between species may depend on the shared mutualistic partners' traits, especially traits related to their ability to attract mutualists. In any mutualistic system, when high-quality partners exhibit highly competitive ability, we can expect that sharing these high-quality partners will play a disproportionate role in shaping indirect effects among mutualistic species within a community (Johnson & Bronstein, 2019).

The probability and number of plant interactions with dominant and subordinate ants were not influenced by the distance between plants with higher promoter degree values, their neighbouring or non-neighbouring plants. A recent experimental study found that EFN-bearing tree species can increase ant biomass in themselves and adjacent non-EFN trees (Staab et al., 2023). In other plant–animal mutualisms (e.g., pollination and dispersion mutualisms), more attractive plants facilitate the visitation of shared partners to less attractive neighbouring species (Carlo, 2005; Johnson et al., 2003; Seifan et al., 2014). However, competition can occur when less attractive plant species are visited less when they are close to more attractive species (Carlo et al., 2003; Yang et al., 2011). This apparent contradiction improves our ability to generalize the factors driving indirect effects among mutualistic species. For instance, indirect effects among mutualistic species likely depend on how shared partner species explore resources in the habitat. Unlike bees interacting with plants in pollination systems, which can forage further than a kilometre from the nest (Kendall et al., 2022; Schneider, 1989; Waddington et al., 1994), ants have limited foraging ranges. Even the dominant ants in our study sites normally forage less than 6 m from the nest (Baccaro & Ferraz, 2013). Thus, although the promoter plants are the most attractive to ants, their impact on neighbouring plants can be limited. In this case, the effect of sharing ants between neighbouring plants would not necessarily be weak but would be much more local than other interactions, emerging as neutral at the community scale.

While we used Müller's index to measure indirect effects of length 2 (Plant 1 → Ant → Plant 2), we acknowledge the limitation of this approach in capturing longer chains of indirect effects. Longer chains (such as Plant 1 → Ant 1 → Plant 2 → Ant 2 → Plant 3) can occur and may significantly influence the dynamics of the studied system. These longer indirect effects highlight how changes in the interaction frequency of one plant species can cascade through the network, affecting other plant species in more complex ways (Menge, 1996). Future studies should consider employing methods that can capture these longer chains of indirect effects, such as using the powers of the adjacency matrix to analyse the extended network of indirect effects among co-occurring plants with EFNs.

4.2 | Impact of indirect effects on leaf herbivory patterns

Our results show that although Bignonieae species with a higher promoter degree did monopolize visits of shared ants, mainly the dominant ones, this did not influence the patterns of herbivory on these plants nor other receptor plants in the community. Likewise, herbivory levels were similar and low across all plant individuals irrespective of the distance from the plant species with the highest promoter degree. Considering that EFN-bearing plants in our plots have the same probability of interacting with dominant or subordinate ants and are under low herbivory pressure, any changes in the patterns of ant attendance due to indirect effects could have a relatively low impact on herbivory patterns (but see Staab et al., 2023), regardless of their promoter/receptor degree. Therefore, in this case, indirect interactions via shared ants are neutral (e.g., Armbruster & McGuire, 1991; Caruso, 1999; Tur et al., 2016).

These results might suggest that the consequences of indirect effects mediated by ants on EFN-bearing plants are relatively minor

in nature. Alternatively, they may be challenging to detect due to the involvement of a highly variable third agent, herbivores, which introduces a dependence on a specific ecological context. For example, many herbivores attack these plants, some of which are specialists (Fine et al., 2013). Thus, if individuals from a dominant ant species visit several plant species, they will encounter herbivores with distinct behavioural traits (Figure S6). While some herbivores are negatively affected by ants, others may have effective escape and camouflage strategies that allow them to persist (Itioka et al., 2000). On the other hand, less attractive plants monopolized by subordinate ants may encounter herbivores with little ability to escape. Although less aggressive, in some cases, subordinate ant species may repel some of these herbivores (Leal et al., 2006), decreasing herbivory in particular plant species. Another point is that plants have an arsenal of defensive traits beyond protective ants, including tolerance and escape strategies (Boege & Marquis, 2005). Therefore, herbivores might be more challenging to predict by the indirect effects via ant sharing in this system, given that other defence mechanisms (e.g., secondary metabolites) also modify patterns of herbivory (Coley & Barone, 1996; Crawley, 1993; War et al., 2012). These factors, combined with the complex interactions between ants, herbivores, and plants, may contribute to the difficulty of predicting and detecting the indirect effects of ant sharing in this system. Instead, it appears that the direct effects of ants in reducing herbivory are more critical in this particular system (Rosumek et al., 2009; Trager et al., 2010).

A potential caveat to our conclusions is that we did not quantify carbohydrate resources other than Bignonieae EFN that ants could have used. Although we never observed other significant resources used by local ant species during our sampling, more minor food sources and other (non-Bignonieae) plants with active EFNs must have occurred in at least some plots. However, these resources are unlikely to have influenced our results. In fact, we found that the most attractive Bignonieae species were monopolized by the ant assemblage, especially the dominant ant species in our study site. This pattern indicates that, despite alternative resources, they do not prevent monopolizing the more relatively valuable Bignonieae species by the most competitive ant species occurring in the community.

To the best of our knowledge, this is the first community-level study focused on understanding the indirect effects on protection mutualisms between EFN-bearing plants and ants, exploring whether such indirect effects modify ant protection efficiency across plant species. Our results indicated that indirect effects among plant species with EFNs sharing protective ants play a minor role in the outcome of this mutualistic interaction for the plants. This result is quite different from other mutualisms investigated so far. For example, studies of hemipteran-ant protection mutualisms (Delabie, 2001; Rico-Gray & Oliveira, 2007) have shown that aphids can compete for ant protection with other aphid species (Addicott, 1978) as well as with EFN-bearing plants (Engel et al., 2001; Katayama et al., 2013). As aphid colonies grow, they become a greater reward for ants, influencing their foraging patterns (Katayama et al., 2013). This shift in ant visitation to the most attractive and larger aphid colonies generates negative indirect effects (such as those proposed by our *competition hypothesis*) on other aphid colonies and/or EFN-bearing plants over time (Buckley, 1987). Strong indirect effects have also been noted in pollination mutualisms, with species traits determining the outcome of indirect interactions between plant species sharing pollinators (Bergamo et al., 2017, 2021). In those systems, plant species with more flowers can attract shared pollinators (Ghazoul, 2006; Rathcke, 1983), with results consistent with a competition scenario (Bergamo et al., 2021). Therefore, overall, indirect effects are not a universal property of mutualism and can depend on the intrinsic aspects mediating the net benefit of partners and on the mutualistic partners' life history.

AUTHOR CONTRIBUTIONS

Caroline Souza, Anselmo Nogueira, and Pedro J. Bergamo conceived the idea; Anselmo Nogueira and Fabrício B. Baccaro collected the data; Caroline Souza and Anselmo Nogueira analysed the data and led the writing of the manuscript; Anselmo Nogueira provided guidance and oversight for the article; Laura C. Leal offered support during the process. Caroline Souza, Laura C. Leal, Fabrício B. Baccaro, Pedro J. Bergamo, Judith L. Bronstein, Pedro J. Rey, and Anselmo Nogueira contributed to the writing and gave final approval for publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data are available on FigShare: https://doi.org/10.6084/m9.figshare.26496472.v2 (Souza et al., 2024).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1. Model comparison describing the variation of promoter and receptor degree of plant species considering the entire ant assemblage related to the average number of active nodes (NAN) and the secretory area of extrafloral nectaries (SA).

Table S2. Model comparison describing the variation of promoter and receptor degree of plant species partitioned by the dominant ant

species related to the average number of active nodes (NAN) and the secretory area of extrafloral nectaries (SA).

Table S3. Model comparison describing the variation of promoter and receptor degree of plant species partitioned by the subordinate ant species related to the average number of active nodes (NAN) and secretory area of extrafloral nectaries (SA).

Figure S1. Ant recruitment on plants (median) shows that dominant ant species have significantly higher average recruitment per active EFN than subordinate ants.

Figure S2. Absence of a relationship between the number of active nodes and the receptor degree of plant species relative to (a) the entire ant assemblage, (b) dominant ant species partition, and (c) the subordinate ant species partition.

Figure S3. Partial plots showing the proportion of leaflets consumed and the promoter degree concerning (a) all shared ants and (b) the shared dominant ant species; likewise, partial plots showing the average percentage of herbivory and (c) receptor degree for all visiting ant species and (d) the promoter degree for subordinate ant species.

Figure S4. (a) The probability of interaction and (b) the number of dominant and subordinate ant species feeding on EFN of Bignonieae plants did not differ among the promoter plant species, neighbour plants, and non-neighbour plants in the plot scale evaluation.

Figure S5. Plot showing that the mean percentage of damaged leaf area per plant did not differ between promoter, neighbour, and non-neighbour plants.

Figure S6. The feeding strategies and ability to evade ant predation vary among herbivores.

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